

Session 5 · Deck 5.B5

Determining the Coefficients

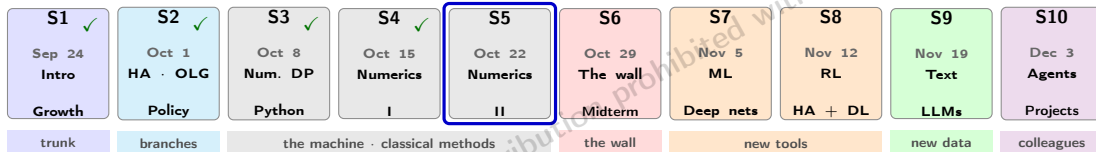
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Where We Are

you are here: 5.B5 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 5.B1 the residual · 5.B2 choosing a basis · 5.B3 Chebyshev · 5.B4 finite elements · **5.B5** determining the coefficients · 5.B6 the worked example.

Weighted Residuals

The standard answer to “what does small mean” is: **zero in a weighted integral sense**. Choose $n + 1$ weight functions $w_i : \Omega \rightarrow \mathbb{R}^m$ and require

$$\int_{\Omega} w_i(x) R(x; \theta) dx = 0, \quad i = 0, \dots, n.$$

That is $n + 1$ equations for $n + 1$ coefficients — square, and generically solvable.

This is why projection methods are also called **weighted residual methods**. Every named method in the literature is one choice of $\{w_i\}$.

And the payoff: an **intractable functional equation** has become a **standard nonlinear system**. Newton for small problems, conjugate gradient for large ones — all of 4.A1 and 4.A3, unchanged.

Notation: the literature writes the weights as f_i and the basis as ψ_i , g_i or φ_i depending on the author. Here the basis is always ψ_i and the weights always w_i .

What the Weight Is Actually For

A policy function does not misbehave uniformly. It may bend sharply near a constraint and be almost linear elsewhere — so a residual of a given size means different things in different regions.

The weight lets you say where accuracy matters: put more weight where the function varies rapidly, and the solve will spend its coefficients there.

The source's analogy is apt: the weight handles “**heteroskedasticity**” in the function being approximated. It is the same instinct as weighted least squares, and the same instinct as 3.A2's log-spaced asset grid.

Quant _ Macro Lec. 9, “Weight function” (both frames).

Basis functions as weights

A natural choice is to use the **basis itself**. If the basis was chosen to match the features of the function, it already puts its variation where the function's is — so weighting by it emphasises exactly the right regions. That choice has a name: **Galerkin**.

I. Least Squares

Take $w_i(x) = \frac{\partial R(x; \theta)}{\partial \theta_i}$.

This is not arbitrary — it is the first-order condition of the variational problem

$$\min_{\theta} \int_{\Omega} R^2(x; \theta) dx,$$

whose FOC is exactly $\int_{\Omega} \partial_{\theta_i} R \cdot R dx = 0$.

So least squares is the **regression** answer: minimise the squared residual over the domain, and you are choosing the weights without realising it.

Quant_Macro Lec. 9, "Weight function I: Least Squares".

Why it is attractive

It always generates **symmetric** matrices — convenient in the proofs, and exploitable by algorithms that are faster and use less memory.

And why it is not: least squares is prone to **ill-conditioning**, and the resulting systems can be awkward to solve. Squaring the residual squares the condition number too.

II. Subdomain, and III. Moments

Subdomain. Split Ω into $n + 1$ pieces Ω_i and take step functions,

$$w_i(x) = \mathbf{1}\{x \in \Omega_i\},$$

so the condition is that the residual **averages to zero on each piece**. It may be large inside a piece, as long as it cancels.

Moments. Take $\{1, x, x^2, \dots, x^n\}$ as weights:

$$\int_{\Omega} x^i R(x; \theta) dx = 0.$$

The first n moments of the residual are zero.

Quant_Macro Lec. 9, "Weight Function II: Subdomain Approach" and "Weight function III: Moments".

Read it carefully

The weights are $\{1, x, x^2, \dots\}$ — the first one is the constant, so the first condition just says the residual *integrates to zero*. The rest set its higher moments to zero.

The verdict on moments is 5.B2's argument again: fine for $n = 2$ or 3 , and for higher orders the monomials are so collinear that rounding error dominates. **"Moments are to be avoided."**

IV. Collocation

Take the weights to be Dirac deltas:

$$w_i(x) = \delta(x - x_i).$$

Since $\int \delta(x - x_i) R(x) dx = R(x_i)$, the conditions collapse to

$$R(x_i; \theta) = 0, \quad i = 0, \dots, n.$$

The residual is exactly zero at $n + 1$ points and unconstrained in between — 5.B1's picture.

The attraction is cost: no integral is ever computed. Each condition is one function evaluation, which matters when R is expensive and strongly nonlinear.

Picking the points

A systematic choice uses a density μ_γ , placing x_j so that $\int_{-1}^{x_j} \mu_\gamma = j/n$. Larger γ clusters points at the endpoints; equi-spacing is the limit $\gamma \rightarrow 0$.

Read the limit carefully: $\gamma = 0$ substituted into μ_γ gives **the zero function**, which is not a density. Equi-spacing is the limit $\gamma \rightarrow 0$, not the value $\gamma = 0$.

Quant_Macro Lec. 9, "Weight function IV: Collocation or pseudospectral" and "Intuition: $\mu_\gamma(x)$ and the Case $\gamma = 0$ ".

Collocation Against Subdomain

	Collocation	Subdomain
Weight	$\delta(x - x_i)$	step function on Ω_i
Residual is	zero at points	zero on average over each piece
Evaluation	pointwise only	an integral per condition
Cost	lower	higher
Best for	strongly nonlinear , spectral bases	smooth , PDE-like problems
Limitation	says nothing between the points	integration overhead

The shared weakness

Neither controls the residual *between* the conditions. Collocation leaves it free between points; subdomain lets it cancel within a piece. That is why you always grade the finished solution on a **different, finer set of points** — 3.A1's separation of grids.

V. Orthogonal Collocation, and VI. Galerkin

Orthogonal collocation. Collocation, with two extra commitments: the basis is a family of **orthogonal polynomials**, and the collocation points are the **roots of the n -th one**.

Those roots are not equi-spaced — they cluster at the endpoints, which is exactly where 5.B3 said an equi-spaced rule fails. The result converges exponentially for smooth functions, and the standing verdict in this literature is “surprisingly good performance”.

Galerkin. Make the residual **orthogonal to every basis function**:

$$\int_{\Omega} \psi_i(x) R(x; \theta) dx = 0.$$

What Galerkin means

The residual has **no component the basis could still represent**. Anything left over is invisible in this function space — so you have extracted everything the basis can give.

If the basis is complete, the Galerkin solution converges **pointwise** to the truth. It is accurate and robust, and *difficult to code*: every condition is an integral of a nonlinear function. Rule of thumb:

Galerkin at order n is about as accurate as collocation at $n + 1$ or $n + 2$.

Measured: Six Weight Functions, One Model, One Basis

The quarterly RBC of Block A, Chebyshev degree 6 in k for each of 7 shock states — **49 unknowns**, identical in every row. Only the weight function changes. All six converge; integrals by 40-node Gauss–Legendre.

Weight	$\max \mathcal{E} $	sec
orthogonal collocation	$10^{-5.78}$	0.11
least squares	$10^{-5.61}$	0.80
Galerkin	$10^{-5.55}$	0.20
moments	$10^{-5.55}$	0.15
collocation, equi-spaced	$10^{-5.39}$	0.11
subdomain	$10^{-5.05}$	0.14

Every method lands within 0.73 of a decade of every other — a factor of 5.4 between best and worst. The choice of weight function is worth less than one extra basis coefficient, which 5.B6 measures at 1.2 decades for two.

And the cheapest is the best: orthogonal collocation wins on accuracy *and* ties for fastest, because it evaluates no integrals at all.

Ledger collocation. Warm-started from the first-order perturbation solution; residual norms $\leq 1.4 \times 10^{-14}$ except least squares, 6.9×10^{-7} .

Two Things the Table Says

1. Moments and Galerkin are the same method here.

Their errors agree to nine decimal places: -5.5515379662 against -5.5515379670 .

That is not luck. $\{1, x, \dots, x^n\}$ and $\{T_0, \dots, T_n\}$ **span the same space**, so “orthogonal to every monomial” and “orthogonal to every Chebyshev polynomial” are the same $n + 1$ linear conditions. In exact arithmetic they are identical.

The warning against moments is therefore **purely numerical**: the conditions are right, and the basis you write them in destroys them. At $n = 6$ that has not bitten yet; 5.B2’s Hilbert table says where it will.

Ledger collocation.rows; full precision in the JSON.

2. Least squares is 7x slower

0.80 s against 0.11 s, for no accuracy gain — and it is the one method that did not drive its residual to machine zero, stopping at 6.9×10^{-7} . Exactly the predicted ill-conditioning, showing up as a solver that will not converge the last six digits.

The source’s “Galerkin $\approx$ collocation at $n+1$ or $n+2$ ” is not what we measure: here *orthogonal* collocation beats Galerkin at the same n .

Takeaways

- “Make the residual small” becomes $\int_{\Omega} w_i R dx = 0$ for $n + 1$ weights — a square **nonlinear system**, solved with the tools of 4.A1 and 4.A3. Every named projection method is one choice of $\{w_i\}$.
- **Least squares** is the regression answer and gives symmetric matrices; it is also the most ill-conditioned, and was measured 7× slower with the worst final residual.
- **Subdomain** zeroes the residual on average over pieces; **moments** zeroes its first n moments and is to be avoided at high order; **collocation** zeroes it at points and computes no integrals at all.
- **Galerkin** makes the residual orthogonal to the basis — nothing is left that the basis could represent — and converges pointwise if the basis is complete.
- Measured on one model with one basis: **all six land within 0.73 of a decade**. The weight function matters less than one extra coefficient.
- **Orthogonal collocation wins on accuracy and ties for fastest**, which is why it is the default in this literature.
- And **moments and Galerkin agreed to nine decimals** — they are the same conditions in two bases. The objection to moments is conditioning, not mathematics.

Bridge: Now Solve the Model

Block B now has both halves: a basis (Chebyshev, on a domain, with clustered nodes) and a projection (orthogonal collocation, because it is cheapest and best).

That is a complete method. The last deck runs it on the model Sessions 3, 4 and 5 have all solved — and then puts every method in this course into one table.

Next (5.B6)

The worked example: the same quarterly RBC by Chebyshev collocation, five steps, warm-started from perturbation. Then **one model, four methods** — and the answer to which one you should reach for.